

B.Ed. 2nd Semester — Course 1.2.3 (1st Half)

Learning and Teaching: Learning Question Bank (2020–2025) with Model Answers

Repeated questions are listed once with all years noted. Group A ≈ 50 words, Group B ≈ 150 words (point form), Group C ≈ 300 words.

Compiled from WBUTTEPA / BSAEU B.Ed. 2nd Semester question papers: 2020, 2021, 2022, 2023, 2024 and 2025.

Most repeated: Constructivism + ZPD · Mental health + teacher's role · Rogers' Self-Concept · Thorndike's laws · Causes of forgetting · Classical vs Operant · Remedial teaching · Bruner · Motivation · Self-efficacy · ZPD · Scaffolding · Shaping

GROUP A — 2 Marks Each (any 5 asked, ~50 words)

1. What is conceptual learning? (2020, 2022)

Conceptual learning is the process of grouping objects, events or ideas into categories on the basis of their common features, and then applying that concept to new examples. For instance, once a child understands the concept of 'bird', she can identify a new bird she has never seen.

2. What is shaping? (2020, 2022)

Shaping is a technique of Skinner's operant conditioning in which a desired behaviour is built step by step by reinforcing successive approximations towards the final goal. For example, a teacher rewards each stage of a child learning to write a letter.

3. What do you mean by 'ZPD'? (2020, 2022, 2023)

The Zone of Proximal Development (Vygotsky) is the gap between what a learner can do independently and what he or she can do with help from a teacher or more capable peer. Teaching is most effective within this zone.

4. What do you mean by scaffolding? (2020, 2022)

Scaffolding is the temporary support given by a teacher or peer (hints, modelling, questioning) to help a learner complete a task beyond his current ability. The support is gradually withdrawn as the learner becomes independent.

5. What is 'self-efficacy'? (2020, 2022, 2025)

Self-efficacy (Bandura) is a person's belief in his or her own ability to perform a task and achieve a goal. It develops through mastery experience, observing others, verbal persuasion and emotional state, and strongly affects effort and persistence.

6. Write any two features of cognitive learning. (2020)

(i) It involves mental processes such as thinking, understanding, memory and problem-solving. (ii) The learner is active and learning occurs through insight and organisation of experience rather than mere trial and error.

7. Write any two importance of co-operative learning. / What is co-operative learning? (2020, 2024)

Co-operative learning is a method in which small groups work together to achieve a common goal. Importance: it develops social skills and teamwork, and improves achievement through sharing of ideas and mutual help.

8. What is collaborative learning? (2023)

Collaborative learning is an approach in which learners work jointly in groups to explore problems, share ideas and build knowledge together, with shared responsibility and less teacher control.

9. Write any two differences between collaborative and co-operative learning. (2021)

Collaborative learning is loosely structured and learner-directed; co-operative learning is highly structured with roles assigned by the teacher. Collaborative focuses on shared knowledge-building, co-operative on

completing a common task.

10. What is brainstorming? (2021, 2023)

Brainstorming is a group technique for generating a large number of ideas on a problem in a free atmosphere, where criticism is not allowed and all ideas are welcomed before they are evaluated.

11. Mention two conditions of classical conditioning. (2021)

(i) The neutral stimulus (CS) must be presented just before or along with the unconditioned stimulus (contiguity). (ii) The pairing must be repeated several times for the conditioned response to form.

12. What is personal motivation? (2021)

Personal motivation is an inner drive that arises from an individual's own needs, interests, curiosity and goals, pushing him to act and learn without external pressure.

13. How can a teacher use 'peer tutoring' as a learning strategy? (2021)

The teacher pairs a capable student (tutor) with a weaker one (tutee), trains the tutor, assigns clear tasks, supervises the sessions and gives feedback. This increases individual attention and understanding for both.

14. Mention any two disadvantages of peer tutoring. (2025)

(i) The peer tutor may give wrong or incomplete explanations. (ii) It may cause discipline problems and the tutor may lose his own learning time.

15. What is procedural learning? (2021, 2025)

Procedural learning is learning 'how to do' something — a skill or sequence of steps such as swimming, typing or solving a sum — which becomes automatic with practice.

16. What is overlearning? (2021)

Overlearning is continued practice or revision of material even after it has been mastered. It strengthens memory and increases retention.

17. How is learning both a process and an outcome? (2025)

As a process, learning is the mental and behavioural activity of acquiring knowledge and skills through experience. As an outcome, it is the relatively permanent change in behaviour or knowledge that results from that activity.

18. State two principles of / write the needs for enrichment programme in school. (2025, 2023)

Principles: it should be flexible and challenging, and learner-centred. Needs: it develops talent and creativity, prevents boredom of bright learners and uses their potential fully.

19. Differentiate between recall and recognition in retrieval of memory. (2025)

Recall is reproducing learned material without any cue (as in essay questions). Recognition is identifying learned material when it is presented among options (as in MCQs) and is easier than recall.

20. Differentiate between positive and negative reinforcement in Skinner's theory. (2025)

Positive reinforcement strengthens behaviour by adding a pleasant stimulus (praise). Negative reinforcement strengthens behaviour by removing an unpleasant stimulus (stopping nagging when homework is done). Neither is punishment.

21. What is achievement motivation? (2023)

Achievement motivation (McClelland) is the need to excel, to attain success and to meet a standard of excellence. High achievers set moderately difficult goals and persist.

22. State two strategies for effective memorization. (2023)

(i) Chunking and use of mnemonics to organise information. (ii) Spaced (distributed) revision with self-testing, along with understanding of the meaning.

23. Write the principles of operant conditioning theory. (2023)

Behaviour is controlled by its consequences. Main principles: reinforcement (positive and negative), punishment, shaping, extinction, schedules of reinforcement and immediate reinforcement.

24. Describe the three stages of representation in Bruner's theory. (2024)

Bruner described three modes of representing knowledge: Enactive (through action), Iconic (through images and pictures) and Symbolic (through language and symbols).

25. Mention two measures taken by teachers for securing mental health of learners. (2024)

(i) Providing a friendly, fear-free classroom atmosphere with respect for every child. (ii) Offering guidance, counselling and co-curricular activities as emotional outlets.

26. What is 'encoding' in remembering? (2024)

Encoding is the first stage of memory in which incoming information is converted into a form (visual, acoustic or semantic) that the brain can store.

27. Give two examples of positive transfer of learning from daily life. (2024)

(i) Knowing how to ride a bicycle helps in learning to ride a motorbike. (ii) Learning arithmetic helps in learning algebra.

28. What is learning by insight? (2024)

Learning by insight (Kohler, Gestalt) is the sudden perception of the relationships in a problem situation leading to its solution, rather than gradual trial and error.

29. What is retroactive inhibition? (2024)

Retroactive inhibition is a type of forgetting in which new learning interferes with and weakens the recall of previously learned material.

30. Mention the different steps of information processing in learning. (2022)

Sensory register, attention and perception, short-term (working) memory, encoding, long-term memory storage and retrieval.

31. What is the SQ4R model? (2022)

SQ4R is a study technique with steps: Survey, Question, Read, Reflect, Recite and Review. It improves comprehension and retention of textual material.

GROUP B — 5 Marks Each (any 3 asked, ~150 words, point form)

1. Discuss the principles of 'Operant Conditioning'. (2021)

B.F. Skinner's theory holds that behaviour is shaped and maintained by its consequences.

- **Reinforcement:** positive (adding a reward) and negative (removing an aversive stimulus) both increase behaviour.
- **Punishment:** reduces the chance of a behaviour recurring.
- **Shaping:** reinforcing successive approximations towards the target behaviour.
- **Schedules:** continuous and partial (fixed/variable ratio and interval) reinforcement.
- **Extinction:** behaviour fades when reinforcement is withdrawn.
- **Immediacy:** reinforcement is most effective when given immediately.
- **Experiment:** a hungry rat in the Skinner box learns to press a lever to get food.

2. Mention the differences between classical conditioning and operant conditioning in learning. (2022, 2024)

Basis	Classical (Pavlov)	Operant (Skinner)
Response	Involuntary, reflexive	Voluntary, emitted
Association	Stimulus with stimulus (CS-UCS)	Response with consequence

Basis	Classical (Pavlov)	Operant (Skinner)
Learner	Passive	Active
Reinforcement	Paired with stimulus, before response	Follows the response
Experiment	Dog and salivation	Rat/pigeon in Skinner box
Key terms	CS, UCS, CR, UCR	Reinforcer, shaping, schedule

3. Write the classroom implications of the theory of conditioning with suitable examples. (2023)

- Use praise, stars and marks as reinforcers (a child praised for neat work keeps it neat).
- Break tasks into small steps and shape behaviour gradually.
- Give immediate feedback and knowledge of results.
- Create pleasant associations with school and subjects to reduce fear (e.g. maths phobia).
- Avoid harsh punishment; use ignoring (extinction) for minor misbehaviour.
- Use drill, programmed instruction and habit formation (punctuality, cleanliness).

4. Discuss, in brief, Bruner's theory of Discovery Learning. (2021, 2023)

Learners discover concepts and principles themselves through exploration rather than being told.

- Active learning, inquiry and problem-solving are central; the teacher is a guide.
- Three modes of representation: Enactive (action), Iconic (images), Symbolic (language).
- Spiral curriculum: topics are revisited at increasing levels of difficulty.
- Role of intuitive thinking and readiness for learning.
- Merits: deeper understanding, intrinsic motivation, better retention and transfer.
- Limitations: time-consuming and needs skilled teachers.

5. Discuss Bandura's Social Cognitive Theory. (2025)

Learning takes place by observing and imitating models in a social setting (Bobo doll experiment).

- **Attention:** the learner notices the model.
- **Retention:** the behaviour is remembered.
- **Reproduction:** the learner is able to perform it.
- **Motivation:** reinforcement makes it worth repeating.
- Reinforcement can be direct, vicarious or self-reinforcement.
- Reciprocal determinism: person, behaviour and environment influence each other; self-efficacy is a key concept.
- Implication: teachers should act as good models and use demonstrations.

6. Write a note on 'cognitive constructivism'. (2020)

Piaget's view: children actively construct knowledge through interaction with the environment.

- Key concepts: schema, assimilation, accommodation and equilibration.
- Four stages: sensorimotor, preoperational, concrete operational, formal operational.
- Implications: activity-based and discovery learning, readiness, attention to individual differences.
- The teacher acts as a facilitator who arranges experiences that create cognitive conflict.

7. Differentiate between collaborative and co-operative learning. (2025)

Basis	Collaborative	Co-operative
Structure	Less structured, learner-controlled	Highly structured, teacher-controlled
Roles	Not fixed	Fixed roles assigned
Goal	Shared knowledge building	Completing a common task
Teacher	Minimal role	Active monitor
Assessment	Group process	Individual and group

8. What is co-operative learning? Discuss briefly the role of a teacher in co-operative learning. (2023)

Co-operative learning is a method in which small, heterogeneous groups work together towards a shared goal. Teacher's role:

- Form balanced groups and assign roles (leader, recorder, reporter).
- Set clear goals and explain tasks and rules.
- Provide materials and monitor the groups; intervene when needed.
- Encourage positive interdependence and individual accountability.
- Evaluate individual and group performance and give feedback.

9. Discuss how peer tutoring can be effectively used to influence learning. (2024)

In peer tutoring a more able student teaches a peer under teacher supervision.

- Select and train tutors; pair students suitably.
- Set clear goals and give structured material.
- Monitor sessions and give feedback; rotate roles.
- Benefits: individual attention, comfortable communication, confidence, deeper learning for the tutor, social skills.
- Limitations: tutor may teach wrongly, discipline problems, time loss.

10. Briefly discuss the process for conducting brainstorming in the classroom. (2025)

Brainstorming (Osborn) is a group technique to generate many ideas.

- Define the problem clearly.
- Form groups and appoint a recorder.
- Announce rules: no criticism, welcome unusual ideas, quantity first, build on others' ideas.
- Let students generate ideas freely within a time limit.
- Record every idea.
- Evaluate, group and select the best ideas.
- Summarise and apply the solution.

11. Explain the different types of learning. (2024)

- Gagne's types: signal learning, stimulus-response, chaining, verbal association, discrimination, concept learning, rule learning and problem-solving.
- Motor learning: skills such as writing, swimming.
- Verbal learning: words, language, poems.
- Conceptual and procedural learning; attitude learning (developing values and interests).
- By method: trial and error, conditioning, insight and observational learning.

12. Discuss the various learning strategies / write down the factors that can help to improve retention of memory. (2024, 2022)

- Learn with understanding rather than by rote.
- Organise material: chunking, outlining, mind maps.
- Use mnemonics and imagery.
- Spaced practice rather than massed practice.
- Recitation, self-testing and overlearning; use SQ4R.
- Whole vs part method as per material.
- Factors: interest, attention, motivation, health, intelligence, emotional state, nature of material and feedback.
- Adequate sleep and rest after learning.

13. What are the objectives of co-curricular activities in schools? (2021)

- All-round development of personality.
- Development of leadership, teamwork and social skills.
- Discovery and nurturing of talents and interests.
- Proper use of leisure time.
- Develops discipline, sportsmanship and democratic values.
- Gives emotional outlet and reduces stress.
- Makes learning enjoyable and strengthens school-community links.

14. Discuss the different areas of guidance services that can be provided in schools. (2024)

- **Educational guidance:** choice of subjects, study habits, solving learning difficulties.
- **Vocational guidance:** career information and selection.
- **Personal-social guidance:** emotional and adjustment problems.
- **Health guidance:** physical and mental well-being.
- **Moral guidance:** values and character building.
- Services include orientation, information, counselling, placement and follow-up.

15. Discuss, in brief, the role of teacher in transmission of motivation. (2020, 2021)

- Create an interesting, stimulating and friendly atmosphere.
- Set clear, achievable goals and explain their purpose.
- Relate lessons to learners' needs, interests and life.
- Use praise, rewards and constructive feedback.
- Use varied methods and teaching aids.
- Encourage healthy competition and co-operation.
- Build self-efficacy through success experiences; model enthusiasm.

16. Discuss, in brief, the determinants of mental health. (2020)

- Heredity and physical health.
- Family environment and parenting style.
- School environment and teacher behaviour.
- Peer group and community.
- Socio-economic conditions.
- Emotional maturity and self-concept.
- Cultural values and life experiences.

17. What measures can a teacher take for securing mental health of learners? (2023, 2025)

- Develop warm, friendly teacher-student relationships.
- Recognise individual differences; avoid comparison and labelling.
- Provide guidance and counselling.
- Organise co-curricular activities, games, yoga and relaxation.
- Use positive discipline instead of harsh punishment.
- Maintain contact with parents.
- Teach life skills such as coping and problem-solving.

18. Discuss the teaching strategies that can enhance transfer of learning among students. (2025)

- Emphasise principles and generalisations, not isolated facts.
- Link topics with each other and with daily life.
- Use varied examples and problems.
- Ensure original learning is thorough and meaningful.
- Teach learning-to-learn strategies.
- Give practice in application.

- Promote problem-solving and critical thinking.

19. Write down the principles of the 'Self Concept Theory' of Carl Rogers. (2020)

- Every individual has an inborn actualising tendency.
- The self develops through interaction with significant others.
- Need for positive regard and self-regard.
- Congruence between real and ideal self gives adjustment; incongruence gives anxiety.
- Conditions of worth distort the self; unconditional positive regard helps growth.
- Goal: the fully functioning person.

20. Short note: Remedial Teaching. (2021, 2023, 2025)

Remedial teaching is corrective instruction for students who have not reached the expected level.

- Steps: identify the difficulty (diagnostic test, observation), find causes, plan, teach, re-test and follow up.
- Strategies: individual attention, peer tutoring, simple language, activity methods, graded worksheets, extra practice.
- Encouragement and positive reinforcement; involve parents.

21. Discuss briefly the causes of forgetting. (2020, 2022)

- Decay of memory traces through disuse.
- Interference: proactive and retroactive.
- Repression of painful experiences (Freud).
- Lack of interest, attention or motivation.
- Poor original learning and rote memorisation.
- Retrieval failure due to missing cues.
- Physical and emotional factors: fatigue, stress, illness.
- Nature of material: lengthy or meaningless matter is forgotten faster.

22. Discuss briefly the educational implications of Thorndike's major laws of learning. (2022)

- Law of Readiness: prepare and motivate learners before teaching.
- Law of Exercise: provide drill, revision and practice.
- Law of Effect: use praise and reward; avoid failure-heavy tasks.
- Teach from known to unknown with meaningful association.
- Create interest and a positive attitude.

23. Discuss briefly the educational significance of Gestalt theory. (2022)

- Present the topic as a whole first, then in parts.
- Encourage problem-solving and insight.
- Organise material in meaningful patterns.
- Use projects and activities; allow time for insight.
- Relate new learning to past experience.

GROUP C — 10 Marks Each (any 1 asked, ~300 words)

1. What is constructivism? Briefly explain the 'Zone of Proximal Development' in Vygotsky's theory of learning. (3+7) (2020, 2022, 2024)

Constructivism (3 marks)

Constructivism is a theory of learning which holds that learners actively construct knowledge from their experiences and prior understanding rather than passively receiving it. Piaget emphasised individual (cognitive) construction, while Vygotsky emphasised social construction. The teacher acts as a facilitator, and learning is active, contextual and collaborative.

Zone of Proximal Development (7 marks)

ZPD is the distance between a child's actual developmental level (what he can do independently) and his potential level (what he can do with guidance from a More Knowledgeable Other).

- **Lower limit:** tasks the child can already perform alone. **Upper limit:** tasks possible only with help.
- **Scaffolding:** temporary support such as hints, modelling and questioning, withdrawn gradually as competence grows.
- **Example:** a child cannot solve a quadratic equation alone but can with the teacher's hints; later she solves it independently.

Educational implications

- Use group work and peer tutoring; assess potential as well as performance.
- Plan graded tasks slightly above the learner's present level.
- Teacher acts as guide; use reciprocal teaching and dialogue.

2. What do you understand by mental health? How can a teacher help in fostering the mental health of students? (3+7) (2020, 2022)

Meaning (3 marks)

Mental health is a state of emotional, psychological and social well-being in which a person can cope with the normal stresses of life, work productively, adjust with others and realise his potential. It is not merely the absence of mental illness.

Role of the teacher (7 marks)

- Create a warm, democratic and fear-free classroom climate.
- Treat every child with respect; avoid labelling, comparison and humiliation.
- Identify early signs such as withdrawal, aggression, anxiety or falling grades, and counsel or refer.
- Provide co-curricular activities, games, art and music as emotional outlets.
- Set realistic expectations; give praise and constructive feedback to build self-esteem.
- Teach coping and life skills: stress management, problem-solving, empathy.
- Model emotional stability and coordinate with parents and counsellors.

3. Describe briefly Carl Rogers' Self-Concept theory and its educational implications. (7+3) (2023; also 2025 (short note, implications))

Theory (7 marks)

Self-concept is the organised set of perceptions a person holds about 'I' or 'me'. Rogers described three parts: **self-image** (what I am), **self-esteem** (how I value myself) and **ideal self** (what I want to be).

- Every individual has an inborn actualising tendency towards growth.
- The self develops through interaction and the evaluations of significant others.
- Congruence (real self close to ideal self) leads to adjustment; incongruence leads to anxiety and maladjustment.
- Unconditional positive regard promotes a healthy self; conditions of worth distort it.
- The fully functioning person is open to experience, trusting, creative and lives in the present.

Educational implications (3 marks)

- Create learner-centred, non-threatening classrooms; accept students unconditionally.
- Avoid negative labelling; encourage self-evaluation and success experiences.
- Teacher should be empathetic, genuine and act as a facilitator; provide guidance and counselling.

4. Describe with examples Thorndike's major laws of learning and discuss briefly the educational implications of these laws. (5+5) (2024; implications also 2022)

Laws of learning (5 marks)

- **Law of Readiness:** learning is effective when the learner is physically and mentally ready. Example: a hungry child learns to find food faster.
- **Law of Exercise:** connections are strengthened by use (practice) and weakened by disuse. Example: multiplication tables, typing.

- **Law of Effect:** a response followed by satisfaction is strengthened; one followed by annoyance is weakened. Example: praise increases effort.
- Secondary laws: multiple response, set or attitude, prepotency of elements, response by analogy, associative shifting.

Educational implications (5 marks)

- Motivate and prepare students before teaching.
- Give regular drill, revision and practice.
- Use praise, rewards and success experiences; avoid punishment and repeated failure.
- Teach from known to unknown with meaningful association.
- Create interest and a good attitude towards learning.

5. State in brief the 'Gestalt Theory of Learning'. Write down the application of Gestalt theory in classroom teaching. (6+4) (2021; significance also 2022)

Theory (6 marks)

Propounded by Wertheimer, Koffka and Kohler. Its central idea is that 'the whole is greater than the sum of its parts'. Learning is by **insight**, the sudden perception of relationships in a problem situation. Kohler's experiment with the chimpanzee Sultan, who joined two sticks to reach a banana, is the classic example.

- Perceptual organisation: figure-ground, proximity, similarity, closure and continuity.
- Insight is not trial and error; it depends on the situation as a whole.
- Insight once gained can be transferred to new situations.

Classroom application (4 marks)

- Present the topic as a whole first, then break into parts.
- Encourage problem-solving and discovery; give time for insight.
- Organise material in meaningful patterns; relate to past experience; use projects.

6. State in brief the strategies for remedial teaching and enrichment programme taken by schools for slow learners. (5+5) (2021; remedial teaching also 2023, 2025)

Remedial teaching (5 marks)

Remedial teaching is special corrective instruction for students who fail to reach the expected level.

- Identify the difficulty through diagnostic tests and observation; find its causes.
- Plan and teach individually or in small groups using simple methods, extra practice and teaching aids.
- Use peer tutoring, activity and play-way methods, graded worksheets and extra classes.
- Re-test, follow up, give encouragement and involve parents.

Enrichment programme (5 marks)

Enrichment broadens and deepens learning, especially for gifted or bright students.

- Strategies: extra reading, projects, quizzes, Olympiads, clubs, creative writing, field trips, independent study and mentoring.
- Needs: develops talent and creativity, prevents boredom, uses potential fully.
- Principles: flexible, challenging, learner-centred and linked to the curriculum.

7. What is forgetting? Discuss the various causes of forgetting. (2+8) (2025; causes also 2020, 2022)

Meaning (2 marks)

Forgetting is the inability to recall or recognise previously learned material when it is needed.

Causes (8 marks)

- **Decay (disuse):** memory traces fade with time.
- **Interference:** proactive (old learning blocks new) and retroactive (new learning blocks old).
- **Repression:** painful memories are pushed into the unconscious (Freud).
- **Lack of motivation and attention:** what is not attended to is not stored.
- **Faulty learning:** rote learning, no understanding, insufficient practice.

- **Retrieval failure:** cue-dependent forgetting; the memory exists but cannot be accessed.
- **Physical and emotional factors:** fatigue, illness, stress, brain injury.
- **Nature of material:** meaningless, lengthy or poorly organised matter is forgotten quickly (Ebbinghaus curve).

8. Write down in brief the theories of transfer of learning. What steps should a teacher follow in class for better transfer of learning? (7+3) (2023; strategies also 2025)

Theories of transfer (7 marks)

- **Mental discipline (faculty) theory:** training of mental faculties like memory and reasoning transfers to all subjects.
- **Identical elements theory (Thorndike):** transfer occurs to the extent that two tasks share common elements.
- **Generalisation theory (Judd):** principles learned in one situation are applied in others.
- **Transposition theory (Gestalt):** insight into relationships carries over to new situations.
- **Two-factor theory (Spearman) and Bagley's theory of ideals:** general ability and ideals or attitudes transfer widely.

Steps for the teacher (3 marks)

- Teach concepts and principles rather than isolated facts; show links between subjects.
- Give varied examples and practice in application; ensure thorough original learning.
- Motivate students and develop positive attitudes.